

National Institute of Technology Warangal

Software Engineering Lab

Classroom & Seminar Hall Booking System

Vision Document

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1. Introduction

This document defines the vision and scope of the Classroom and Seminar Hall Booking System (CSHBS), developed as part of the Software Engineering Lab.

1.1 Purpose

To provide a clear understanding of system objectives, scope, and stakeholder expectations.

1.2 Scope

CSHBS is a web-based application that enables booking and management of classrooms and seminar halls by faculty, class representatives, and administrators.

1.3 Definitions, Acronyms, and Abbreviations

- CSHBS – Classroom and Seminar Hall Booking System
- CR – Class Representative
- Admin – Infrastructure Administrator
- Slot – Time interval for booking

1.4 References

SE Lab Guidelines, IEEE Vision Document Template.

1.5 Overview

This document covers positioning, stakeholders, product overview, features, constraints, quality attributes, and priorities.

2. Positioning

2.1 Business Opportunity

Manual classroom scheduling leads to conflicts and inefficient resource usage. A centralized digital system improves transparency and utilization.

2.2 Problem Statement

The problem of unmanaged room scheduling affects faculty, students, and administrators the impact of which is double booking, wasted resources, and disruptions a successful solution would be real-time availability, authorization control, and notifications.

2.3 Product Position Statement

For academic institutions requiring efficient room management, CSHBS is a web-based booking system that provides real-time scheduling and approval workflows unlike manual coordination methods.

3. Stakeholder and User Descriptions

3.1 Market Demographics

Target users include faculty, students, and administrators in medium to large educational institutions.

3.2 Stakeholder Summary

Name	Description	Responsibilities
Faculty	Teaching staff	Book classrooms and seminar halls
CR	Student representative	Request rooms for class needs
Admin	Infrastructure authority	Approvals and monitoring
Project Team	SE students	Development and testing

3.3 User Summary

User	Description	Responsibilities	Stakeholder
Faculty	Course instructors	Book and cancel rooms	Academic Dept
CR	Class Representative	Request bookings	Students
Admin	System manager	Approvals, reports	Institute

3.4 User Environment

Users access the system via web browsers on desktops or mobile devices.

3.5 Key Stakeholder or User Needs

Need	Priority	Current Solution	Proposed Solution
Avoid conflicts	High	Manual checking	Automated validation
Room availability	High	Notice boards	Live dashboard
Authorization	High	Verbal approval	Role-based approval
Notifications	Medium	Informal messages	System alerts

3.6 Alternatives and Competition

Alternative	Strengths	Weaknesses
Manual Registers	Simple	Error-prone
WhatsApp Groups	Fast	No validation
Generic Calendars	Ready-made	No approvals

4. Product Overview

4.1 Product Perspective

CSHBS is a standalone web application with role-based access.

4.2 Summary of Capabilities

Capability	Description
Room Booking	Book classrooms and seminar halls
Availability Check	Real-time availability display
Approval Workflow	Admin authorization
Notifications	Booking confirmations
Reports	Usage history

4.3 Assumptions and Dependencies

Assumes internet access and authenticated institutional users.

5. Product Features

This section describes the primary features offered by the Classroom and Seminar Hall Booking System. These features are designed to address the needs of stakeholders and ensure efficient utilization of academic infrastructure.

5.1 Core Features

- The system provides secure authentication with role-based access for faculty, class representatives, and administrators.
- Faculty members can book classrooms for regular or extra classes by selecting date, time slot, and room based on availability.
- Class representatives are allowed to request classrooms or seminar halls for academic activities with proper authorization.
- The system displays real-time availability of classrooms and seminar halls to prevent double booking.
- Seminar halls can be booked for events, guest lectures, workshops, and academic sessions.
- An approval workflow ensures that all booking requests are verified by the respective managing authority.
- Users can modify or cancel bookings within permitted time limits.
- The system sends booking confirmation and status notifications to users through system alerts and optional email or SMS services.
- Administrators are provided with a dashboard to monitor bookings, manage rooms, and generate usage reports.

6. Constraints

The following constraints apply to the development and operation of the Classroom and Seminar Hall Booking System.

- The project must be completed within a single academic semester as part of the Software Engineering Lab.
- The development team consists of only three students, limiting available development resources.
- The project has no allocated budget and must rely on open-source technologies.
- The system must operate on standard institutional infrastructure without specialized hardware.
- Minimal user training should be required for faculty and students.

- The system must enforce role-based access control to ensure security and data integrity.

7. Quality Ranges

This section defines the expected quality attributes of the system and their acceptable ranges.

- The system should respond to user actions within 2 seconds under normal load.
- The system should maintain at least 95% uptime during institutional working hours.
- Users should be able to complete a booking process within one minute.
- The system should support multiple concurrent users during peak hours.
- Security mechanisms should prevent unauthorized access and booking manipulation.
- The system should be maintainable and extensible for future enhancements.

8. Precedence and Priority

The features of the Classroom and Seminar Hall Booking System are prioritized as follows.

8.1 High Priority (Must Have – Phase 1)

- User authentication and role-based access control
- Classroom and seminar hall booking functionality
- Real-time availability checking
- Conflict prevention and double booking control
- Admin approval workflow

8.2 Medium Priority (Should Have – Phase 2)

- Notification system (email or SMS)
- Booking modification and cancellation
- Booking history and reports

8.3 Low Priority (Future Enhancements)

- Mobile application support
- Integration with academic timetables
- Advanced analytics and optimization features

9. Other Product Requirements

9.1 Applicable Standards

The system should comply with standard web technologies, security guidelines, and software engineering best practices.

9.2 System Requirements

The system requires a web server, database support, and modern web browsers on the client side.

9.3 Performance Requirements

The system should efficiently handle concurrent booking requests with minimal latency.

9.4 Environmental Requirements

The system should operate reliably within the institutional network and be accessible from both desktop and mobile devices.

10. Documentation Requirements

10.1 User Manual

A user manual must be provided explaining system usage for faculty, class representatives, and administrators.

10.2 Online Help

The system should include context-sensitive help and frequently asked questions.

10.3 Installation and Configuration

Technical documentation should describe system installation, configuration, and deployment procedures.

10.4 Labeling

All documents should include version information, project title, and authorship details.

Appendix A: Feature Attributes

This appendix describes the attributes associated with the key features of the Classroom and Seminar Hall Booking System. These attributes help in understanding the importance, effort, risk, and stability of each feature during development.

Feature	Status	Benefit	Effort	Risk
User Authentication	Approved	Critical	Medium	Low
Classroom Booking	Approved	Critical	Medium	Low
Seminar Hall Booking	Approved	Critical	Medium	Low
Real-Time Availability	Approved	Important	Medium	Medium
Approval Workflow	Approved	Critical	High	Medium
Booking Modification	Proposed	Useful	Low	Low
Notifications	Proposed	Important	Low	Low
Admin Dashboard	Approved	Important	Medium	Medium
Reports and History	Proposed	Useful	Low	Low